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PROJECT REPORT

AI-BASED SMART STUDY PLANNER WITH PROGRESS TRACKING

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INTRODUCTION

Time management is a major challenge for students, especially when handling multiple subjects, deadlines, and varying difficulty levels. Traditional study methods often rely on static schedules that fail to adapt to real-time changes.

This project presents a Smart Study Planner, which helps students organize their study tasks efficiently by generating a dynamic schedule based on deadlines and difficulty levels. The system also tracks task completion, encouraging better productivity and discipline.

PROBLEM STATEMENT

Students often struggle to:

- Prioritize tasks effectively
- Manage time across multiple subjects
- Adjust schedules when tasks are delayed

Existing methods like manual timetables are rigid and inefficient.

Objective:

To design a system that automatically generates a prioritized study plan and allows users to track and update their progress dynamically.

OBJECTIVES

- To develop a system that accepts study tasks as input
- To prioritize tasks based on urgency and difficulty
- To generate a structured study schedule
- To allow users to update task completion status
- To apply basic AI concepts like heuristic-based decision making

METHODOLOGY

The project uses a heuristic-based approach to prioritize tasks.

Priority Calculation:

A priority score is calculated using:

Priority = Difficulty / Days Remaining

- Higher difficulty → higher priority
- Fewer days left → higher priority

Scheduling Strategy:

- Tasks are sorted based on priority
- Higher priority tasks are scheduled earlier
- The system generates a simple day-wise plan

Progress Tracking:

- Each task is assigned a status: Pending or Completed
- Users can update task status through the menu system

This approach reflects basic concepts of:

- Intelligent decision-making
- Optimization
- Heuristic search

SYSTEM DESIGN

Modules:

- Task Input Module: Accepts task name, deadline, and difficulty
- Scheduler Module: Calculates priority
- Sorts tasks
- Display Module: Shows structured study plan
- Progress Tracking Module: Updates task completion status
- Menu System: Provides user interaction

IMPLEMENTATION

The project is implemented using:

- Programming Language: Python
  - Libraries Used: datetime (for date calculations)

Key Features:

- Menu-driven interface
- Task prioritization
- Dynamic scheduling
- Task status updates

RESULTS

The system successfully:

- Generates a prioritized study plan
- Helps users focus on urgent tasks
- Allows real-time updates to task status

Example Output:

- Tasks with closer deadlines and higher difficulty are scheduled first
- Completed tasks are updated and reflected in the plan

CHALLENGES FACED

- Handling date calculations correctly
- Designing a simple yet effective priority function
- Maintaining user-friendly interaction in a command-line interface

APPLICATIONS

- Student time management
- Exam preparation planning
- Personal productivity improvement

LIMITATIONS

- No graphical user interface
- No data persistence (tasks are not saved after program ends)
- Scheduling is basic and does not consider daily time limits

FUTURE ENHANCEMENTS

- Add GUI using Streamlit or Tkinter
- Store data using a database
- Use machine learning to predict study time
- Add notifications and reminders

REFERENCES

- Course materials on AI and ML
- Python documentation
- Online resources for scheduling algorithms

CONCLUSION

The Smart Study Planner demonstrates how basic AI concepts like heuristic decision-making can be applied to solve real-world problems. The system improves task prioritization and helps students manage their time more effectively.

Although simple, the project highlights the practical application of AI in everyday life and provides a foundation for further improvements.